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Fewer Tokens, Smaller Cache: Reward-Coordinated Efficient Reasoning

Process Reward Signals Jointly Guide Cache Eviction and Generation Length to Halve Inference Cost Without Sacrificing Accuracy

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Large Reasoning Models generate long chain-of-thought traces that strain inference budgets in two entangled ways: the KV-cache grows with context length, and token generation itself consumes compute. Existing compression methods attack only the cache and apply uniform eviction policies across the entire trajectory. This paper shows that a process reward model (PRM) naturally tracks *where* compression is safe and *when* generation can stop, enabling simultaneous reduction of both costs under a single reward-coordinated framework called RCER.

AT A GLANCE

Problem: LRM inference is expensive in both KV-cache memory and generated tokens; uniform compression sacrifices accuracy.

Why it matters: Efficient LRM deployment is blocked by inference cost; a principled two-sided solution would substantially lower the barrier.

Method: Use a process reward model to guide KV-cache eviction (high-reward steps tolerate more deletion) and generation stopping (stop when the reward signals convergence).

Result: Simultaneous reduction in cache size and token count, with accuracy preserved at compression rates that defeat uniform baselines.

Evidence: Ablations confirm that reward-guided eviction outperforms attention-score and recency heuristics at matched compression ratios.

The Big Picture

Large Reasoning Models (LRMs) are chain-of-thought systems trained to produce extended internal deliberation before outputting a final answer. The deliberation is effective — LRMs achieve state-of-the-art performance on mathematical, scientific, and code reasoning tasks — but it is expensive. A single difficult-problem response may span thousands of tokens of intermediate reasoning, all of which must be stored in the KV-cache during generation and computed token by token.

Two inference costs accumulate jointly: the KV-cache grows linearly with the generated sequence, consuming memory bandwidth and limiting batch size; and the number of generated tokens determines total compute via floating-point operations. These costs are interdependent: a smaller cache means the model has less context, which causes it to regenerate lost information and produce more tokens than it would with a full cache.

This paper identifies the joint nature of the problem and proposes RCER (Reward-Coordinated Efficient Reasoning) as the first method to simultaneously minimize both quantities under a single process reward signal.

Why Existing Approaches Fall Short

Cache-only compression reduces KV-cache size by evicting past tokens based on heuristics: recency (evict oldest tokens), attention score (evict least-attended tokens), or importance scores derived from gradient norms. These approaches are unaware of the reasoning trajectory's progress and evict uniformly, even at critical steps where context loss causes the model to misidentify its reasoning state.

Generation length reduction without cache-awareness typically operates through early-exit mechanisms that terminate generation when some threshold is reached. Without understanding the quality of the reasoning state — only its length — these mechanisms can terminate prematurely, before the model has reached a confident conclusion.

Neither class of method accounts for the *interaction*: compressing the cache causes more generation, and shortening generation can leave the cache unnecessarily large. RCER resolves this by tying both decisions to the same quality signal.

Core Mechanism

RCER augments standard LRM decoding with a process reward model ϕ that is evaluated at each reasoning step k to produce a scalar quality estimate $r_k = \phi(x, t_{\leq k}) \in [0, 1]$, where x is the input problem and $t_{\leq k}$ is the reasoning trace up to step k.

The PRM drives two decisions:

1. **Cache eviction:** When r_k is high (the model has committed to a correct sub-conclusion), the cached representations of older tokens are less necessary — the model's trajectory is converging. RCER evicts an amount proportional to r_k from the oldest entries in the cache.

2. **Generation stopping:** When r_k exceeds a convergence threshold τ and has been stable for δ steps, RCER terminates generation and commits the current answer, avoiding wasteful continuation of a completed reasoning chain.

No additional model beyond the PRM (which may be a lightweight verifier) is required.

Technical Formulation

Let π_θ be the LRM and $r_k = \phi(x, t_{\leq k})$ the PRM score at step k.

Cache eviction budget at step k:

$$b_k = \beta \cdot r_k \cdot |\mathcal{C}_k|$$

where $|\mathcal{C}_k|$ is the current cache size and β is a compression rate hyperparameter. Tokens are evicted from the oldest b_k positions.

Generation stopping criterion:

$$\text{stop if} \quad r_k \geq \tau \quad \text{and} \quad \frac{1}{\delta}\sum_{j=k-\delta}^{k} r_j \geq \tau - \varepsilon$$

This ensures both that the current state is high-quality and that the quality has been stable (not just a local spike).

Effective inference cost:

$$\mathcal{L} = \underbrace{\sum_{k=1}^{K} |\mathcal{C}_k|}_{\text{cache cost}} + \lambda \cdot \underbrace{K}_{\text{token cost}}$$

RCER minimizes $\mathcal{L}$ under the constraint that accuracy remains within ϵ of unconstrained decoding.

Learning or Inference Procedure

RCER is a *training-free* inference-time method. The LRM π_θ and PRM ϕ are both used off-the-shelf without any fine-tuning. The only hyperparameters are:

- $\beta \in [0, 1]$: eviction rate (ablated: 0.1, 0.3, 0.5)
- $\tau \in [0, 1]$: convergence threshold (ablated: 0.7, 0.8, 0.9)
- δ: stability window in steps (ablated: 5, 10, 20)
- λ: weight trading off cache vs. token cost

What the Guarantee Says

No formal guarantee is provided. The empirical claim is that RCER simultaneously reduces both KV-cache occupancy and generated tokens while preserving accuracy within a specified tolerance of unconstrained decoding. The main theoretical insight — that PRM score correlates with the reasoning state's tolerance to context loss — is supported by ablation experiments rather than formal analysis.

Experimental Findings

- **Cache reduction:** RCER reduces peak KV-cache size by 40–55% on GSM8K, MATH-500, and LiveCodeBench at accuracy within 1% of unconstrained decoding.
- **Token reduction:** RCER simultaneously reduces generated tokens by 20–35% through early stopping at convergence.
- **Uniform baselines fail:** At 40% cache compression, attention-score eviction drops accuracy by 4–8%; RCER drops by less than 1%.
- **Interaction confirmed:** Experiments show that cache-only compression at rate β increases token count by 10–25%, confirming the interaction and the necessity of joint coordination.

Ablations and Interpretation

Eviction strategy: Reward-guided eviction consistently outperforms recency, attention-score, and random eviction at matched budgets. The advantage grows with compression ratio, indicating that the reward signal is most critical at high compression where wrong eviction choices compound.

PRM quality: A lightweight 1B-parameter verifier achieves 85–90% of the benefit of a full-scale PRM, suggesting the approach is practical even without access to a high-accuracy reward model.

Stopping ablation: Generation stopping without cache coordination saves tokens but wastes cache — the combined approach is strictly better than either alone on the joint cost metric.

Reference

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